

CISCO CERTIFIED CYBEROPS ASSOCIATE (CBROPS 200-201)

Lab 15.2.7: Packet Tracer — Logging Network Activity

Packet Sniffing, Plaintext Protocol Exploitation, Syslog Analysis & NAT Visibility

COURSE MODULE

Module 15: Network
Monitoring & Tools

CORE PROTOCOLS

FTP (TCP 21), Syslog (UDP
514), ICMP, Telnet

ENVIRONMENT

Cisco Packet Tracer / Multi-
Subnet Topology

KEY CONCEPTS

Traffic Sniffing, NAT
Obfuscation, Log Triage

Addressing Table

DEVICE	INTERFACE	PRIVATE IP ADDRESS	PUBLIC IP ADDRESS	ROLE IN LAB SCENARIO
FTP_Server	Gig0/0	192.168.30.253	209.165.200.227	Enterprise FTP repository storing client files.
SYSLOG_SERVER	FastEthernet0	192.168.11.254	209.165.200.229	Centralized syslog daemon collecting router debug events.
Router2	Serial0/0/1	N/A	209.165.200.226	Core perimeter router generating ICMP debug telemetry.
Sniffer1	Port0 / Port1	N/A (Passive)	N/A (Passive)	Inline packet capturing appliance monitoring Router2 links.
PC-A / PC-B / PC-C	NIC	192.168.1.x / 192.168.2.x	NAT Overload IP	Internal client hosts generating FTP, Telnet & Ping traffic.

Objectives

- **Part 1: Create FTP Traffic** — Activate packet sniffer, establish remote FTP session to 209.165.200.227, authenticate, and perform file transfer (clientinfo.txt).
- **Part 2: Investigate FTP Traffic** — Dissect captured packets on Sniffer1, expose plaintext credential transmission, and define mitigation strategies.
- **Part 3: View Syslog Messages** — Enable debug ip icmp logging on Router2 via Telnet, generate test ICMP traffic across endpoints, and analyze the impact of Network Address Translation (NAT) on security logging.

Part 1: Create FTP Traffic

Step 1: Activate the Sniffing Device

The passive sniffing appliance Sniffer1 is powered on via its **Physical** tab and the sniffing service is enabled in the **GUI** tab. Sniffer1 monitors all frames traversing the link between Router2 and the server subnet.

Step 2: Establish Remote FTP Connection from PC-B

From host PC-B Command Prompt, an FTP session is initiated to the public IP of FTP_Server:

```
PC-B:\> ftp 209.165.200.227
Connected to 209.165.200.227.
220 Service ready for new user.
User (209.165.200.227:(none)): cisco
331 User name okay, need password for cisco.
Password: cisco
230 User logged in, proceed.
```

Step 3: Upload Sensitive Client File (clientinfo.txt)

```
ftp> dir
226 Transfer complete.
ftp> put clientinfo.txt
200 PORT command successful.
150 Opening BINARY mode data connection for clientinfo.txt.
226 Transfer complete.
File transfer in progress: 100% complete.
ftp> dir
-rw-r--r-- 1 ftp ftp 1048 Aug 17 08:30 clientinfo.txt
226 Transfer complete.
ftp> quit
221 Service closing control connection.
```

Part 2: Investigate the FTP Traffic on Sniffer1

Opening the **Sniffer1** GUI tab and inspecting the captured TCP/FTP packet stream reveals the Application Layer control channel commands:

```
Packet #12: FTP Control (Port 21) → Command: USER cisco
Packet #14: FTP Control (Port 21) → Command: PASS cisco
Packet #22: FTP Data (Port 20) → Payload: [CONFIDENTIAL CLIENT RECORDS & ACCOUNT NUMBERS]
```

PART 2 — FTP VULNERABILITY ANALYSIS

What is the security vulnerability presented by FTP?

SOC ANALYST SECURITY FINDING:

The fundamental security vulnerability of the File Transfer Protocol (FTP - RFC 959) is that it **transmits all control and data traffic in unencrypted cleartext**.

- **Credential Sniffing:** Authentication credentials (**USER cisco** and **PASS cisco**) are sent over the wire in plain ASCII with zero cryptographic protection.
- **Confidentiality Breach:** Transferred files (such as **clientinfo.txt**) can be eavesdropped, reconstructed, and stolen by any adversary with access to an inline sniffer, mirrored switchport, or compromised transit router.
- **Data Tampering (MitM):** Cleartext streams lack cryptographic integrity checks, enabling Man-in-the-Middle attackers to modify file payloads in transit without detection.

PART 2 — VULNERABILITY MITIGATION STRATEGY

What should be done to mitigate this vulnerability?

SOC DEFENSIVE ENGINEERING CONTROLS:

1. **Enforce Encrypted File Transfer Protocols:**
 - **SFTP (SSH File Transfer Protocol):** Replaces FTP with a single secure channel running over SSH (TCP Port 22), providing strong public-key/password authentication and symmetric encryption (AES-256-GCM).
 - **FTPS (FTP Secure / FTP over TLS):** Upgrades legacy FTP by wrapping both control (Port 21) and data connections in Transport Layer Security (TLS 1.3).
 - **HTTPS / Secure WebDAV:** Secure RESTful or web-based file management over TLS (TCP Port 443).
2. **Transport Encryption & Segmentation:** If legacy FTP cannot be immediately deprecated, encapsulate all transit traffic through an **IPsec VPN tunnel** or encrypted site-to-site GRE over IPsec.
3. **Access Control Lists (ACLs):** Restrict FTP server access strictly to authorized management IP ranges.

Part 3: View Syslog Messages & Network Visibility Analysis

Step 1: Remotely Enable ICMP Debugging on Router2

From PC-B, Telnet is used to access Router2 and activate real-time ICMP packet debugging:

```
PC-B:\> telnet 209.165.200.226
User Access Verification
Username: ADMIN
Password: CISCO
Router2# debug ip icmp
ICMP packet debugging is on
Router2# logout
```

Step 2: Generate Traffic and Inspect Syslog Server Records

Ping packets are transmitted to Router2 (209.165.200.226) from host PC-B and host PC-A . The SYSLOG_SERVER captures the router's ICMP debug output:

```
SYS-5-CONFIG_I: Configured from console by ADMIN on vty0 (209.165.200.230)
ICMP: echo reply sent, src 209.165.200.226, dst 209.165.200.230, topology BASE, dscp 0 topoid 0
ICMP: echo reply sent, src 209.165.200.226, dst 209.165.200.230, topology BASE, dscp 0 topoid 0
ICMP: echo reply sent, src 209.165.200.226, dst 209.165.200.230, topology BASE, dscp 0 topoid 0
ICMP: echo reply sent, src 209.165.200.226, dst 209.165.200.230, topology BASE, dscp 0 topoid 0
```

PART 3.H — NAT OBFUSCATION & LOG ATTRIBUTION ANALYSIS

Can you tell which echo replies are for PC-A and PC-B from the destination addresses? Explain.

SOC FORENSIC ANALYSIS:

No, it is impossible to distinguish between PC-A and PC-B solely from the destination IP addresses in the syslog messages.

Technical Explanation: Both PC-A and PC-B reside in private IPv4 address spaces (RFC 1918). As their packets traverse the perimeter NAT/PAT gateway (Router1/Edge Router), their private source IP addresses are translated via **Port Address Translation (NAT Overload)** into a single shared public IPv4 address (`209.165.200.230`).

When Router2 generates ICMP echo replies, it addresses them to this shared public NAT address. From the perspective of Router2 and the centralized Syslog server, all ICMP replies share the exact same destination IP (`209.165.200.230`). To correlate which reply corresponds to which physical endpoint, an analyst must correlate router syslog events with the edge gateway's internal NAT translation table (`show ip nat translations`) using port/sequence mappings.

PART 3.I — PC-C PING BEHAVIOR PREDICTION

Ping Router2 from PC-C. What will the destination address for the replies be?

TECHNICAL OBSERVATION:

The destination address for the echo replies generated for PC-C will also be the **public NAT interface address** (`209.165.200.230`) or the specific public NAT pool address assigned to PC-C's gateway subnet).

Because PC-C also resides behind an edge NAT boundary, its private IP address is hidden, and Router2 replies directly to the translated public address visible on the WAN link.

SOC Analyst Architecture Takeaways

- **Legacy Risk** Plaintext protocols (FTP, Telnet, HTTP) must be prohibited across enterprise networks due to passive sniffing vulnerabilities.
- **NAT Blindness** NAT hides internal network host identities from external routers and centralized logs. SOC log ingestion pipelines must ingest both **perimeter NAT flow logs** and **internal DHCP/endpoint logs** to enable IP-to-hostname attribution.
- **Centralized Syslog** Syslog (UDP/TCP 514) provides centralized real-time telemetry. In production, **TLS-encrypted Syslog (RFC 5425)** should be used to prevent tampering with security event logs in transit.